Chapter 1: Exploring the Network



Introduction to Networks

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Chapter 1: Objectives

Upon completion of this chapter you will be able to:

- Explain how networks affect the way we interact, learn, work and play.
- Describe how networks support communication.
- Explain the concept of a converged network.
- Describe the four basic requirements of a reliable network.
- Explain the use of network devices.
- Compare the devices and topologies of a LAN to the devices and topologies of a WAN.
- Explain the basic structure of the Internet.
- Explain how LANs and WANs interconnect to the Internet.
- Describe the impact of BYOD, online collaboration, video, and cloud computing on a business network.
- Explain how networking technologies are changing the home environment.
- Identify some basic security threats and solutions for both small and large networks.
- Explain how the three Cisco enterprise architectures work to meet the needs of the evolving network environment.

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1.1 Globally Connected



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Networking Impacts in Our Daily Lives

- Networks support the way we learn (virtual class rooms)
- Networks support the way we communicate (blogs)
- Networks support the way we work (WebX Conferences)
- Networks support the way we play (online games)

Virtual Classrooms



Online Gaming



Weblog



You can express your thoughts online, share your photos, and join a community of fellow thinkers.



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Networks of Many Sizes



Small Home Networks



Small Office/Home Office Networks



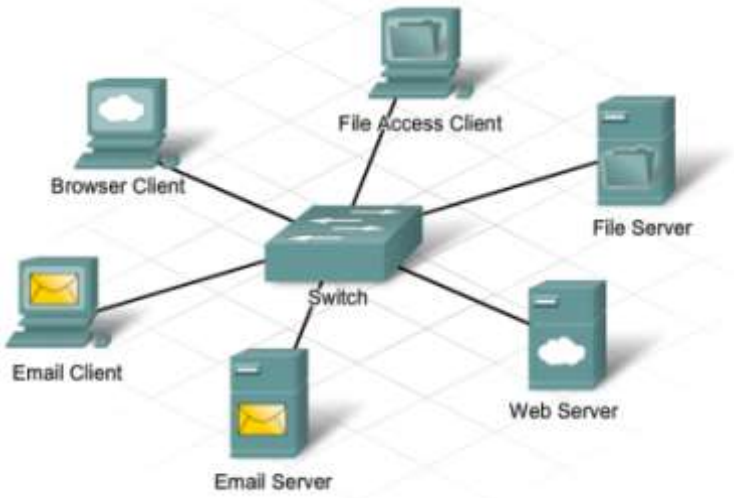
Medium to Large Networks



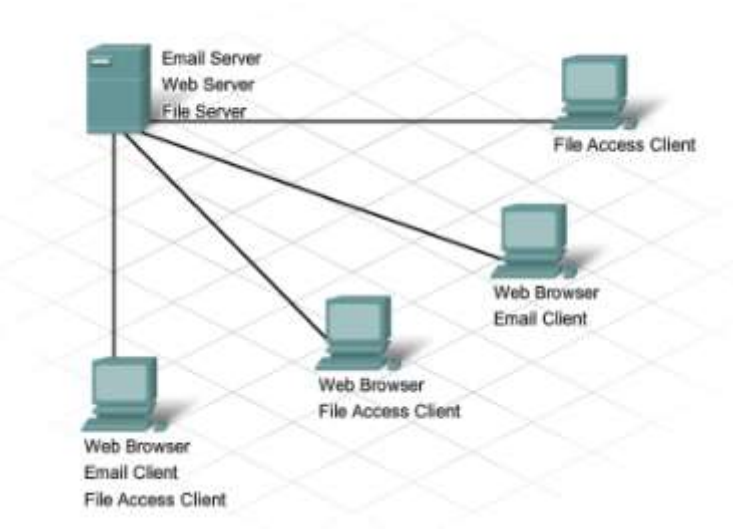
World Wide Networks

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Clients and Servers



Host/End device = client/server/both
Client: Request services
Server: provide information

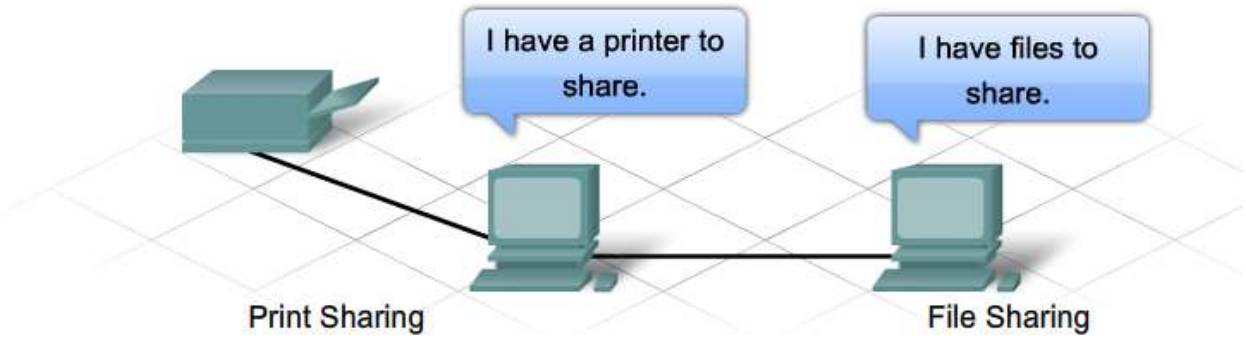


The software on a device determines its role.

- Web browser software like Windows Explorer allows a device to request web pages and to display these pages.
- Email Server software on a device allows a device to store e-mails

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Peer-to-Peer



The advantages of peer-to-peer networking:

- Easy to set up
- Less complexity
- Lower cost since network devices and dedicated servers may not be required
- Can be used for simple tasks such as transferring files and sharing printers

The disadvantages of peer-to-peer networking:

- No centralized administration
- Not as secure
- Not scalable
- All devices may act as both clients and servers which can slow their performance

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1.2 LANs, WANs, and the Internet

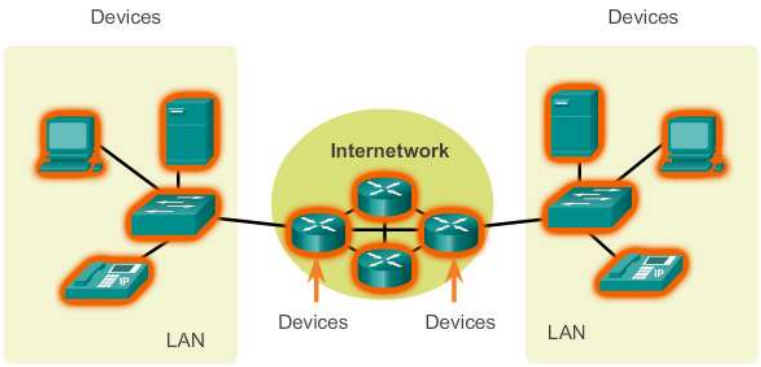


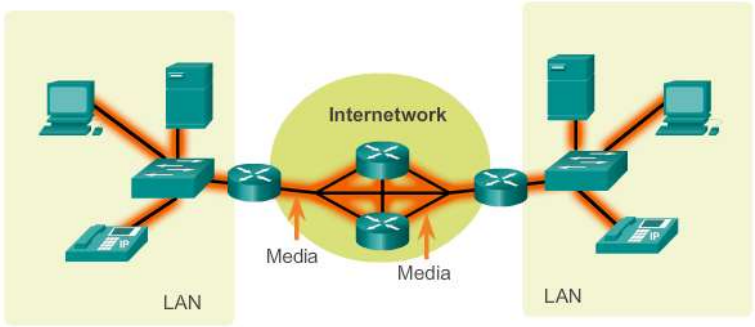
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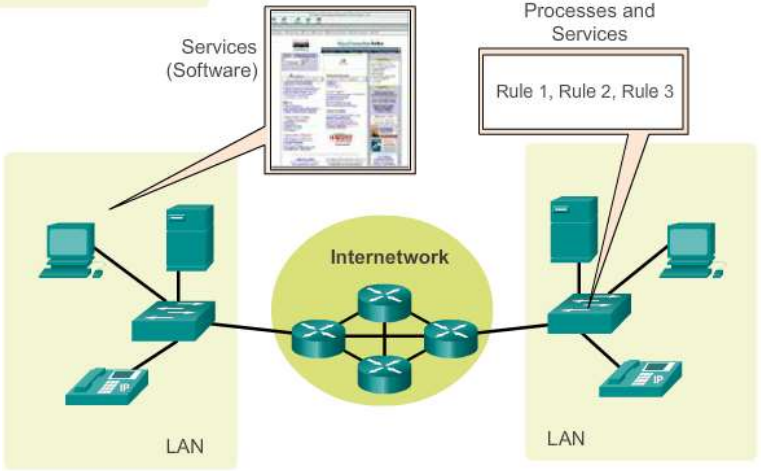
Components of a Network

There are three categories of network components:

- Devices
- Media
- Services







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End Devices

Some examples of end devices are:

- Computers (work stations, laptops, file servers, web servers), network printers, VoIP phones, Security cameras, Mobile handheld devices
- A host device is either the source or destination of a message transmitted over the network, as shown in the animation. In order to distinguish one host from another, each host on a network is identified by an address. When a host initiates communication, it uses the address of the destination host to specify where the message should be sent.

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Network Infrastructure Devices

Examples of intermediary network devices are:

- Network Access Devices (switches, and wireless access points)
- Internetworking Devices (routers)
- Security Devices (firewalls)

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Network Media

electrical impulses

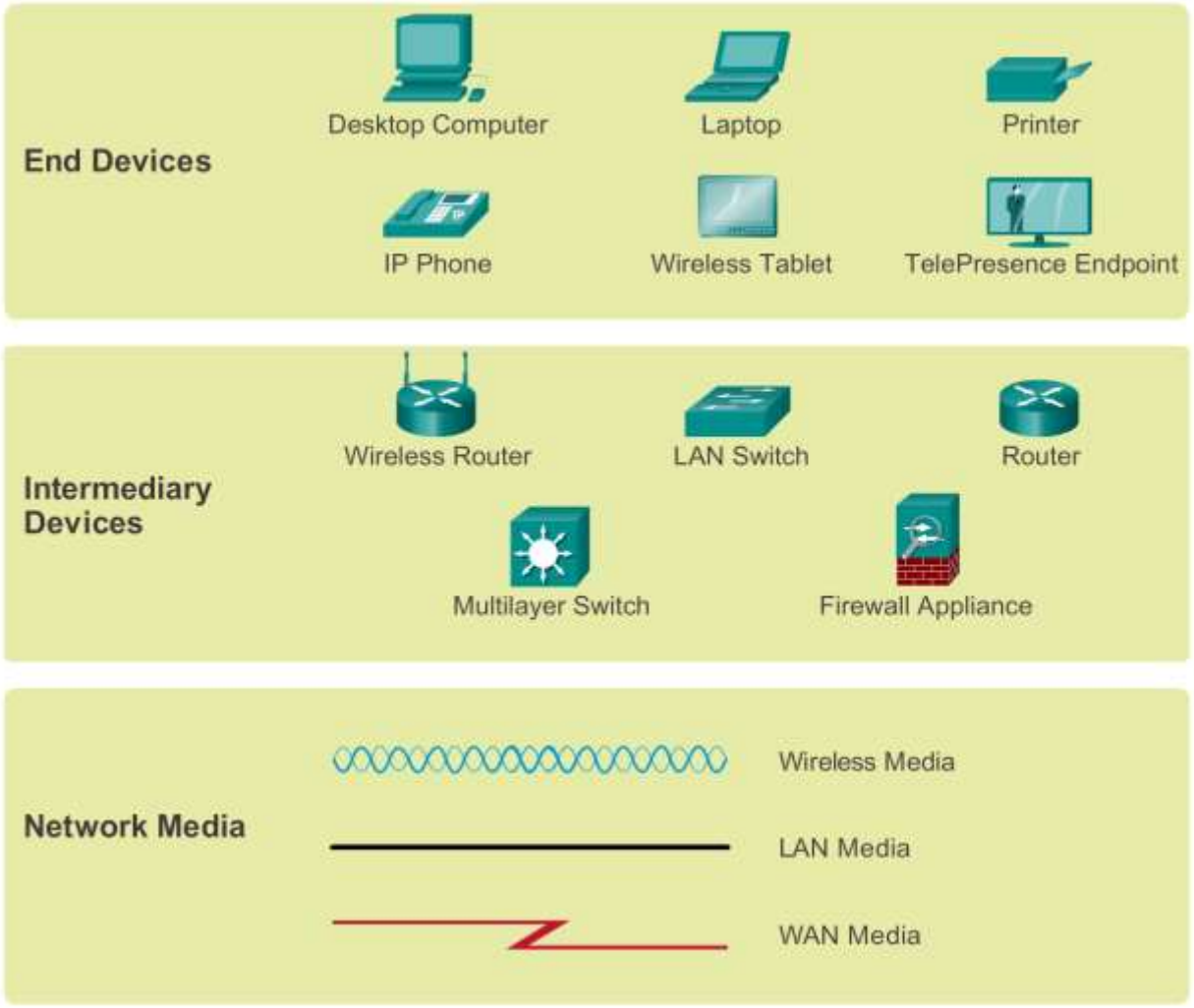
light impulses

electromagnetic waves



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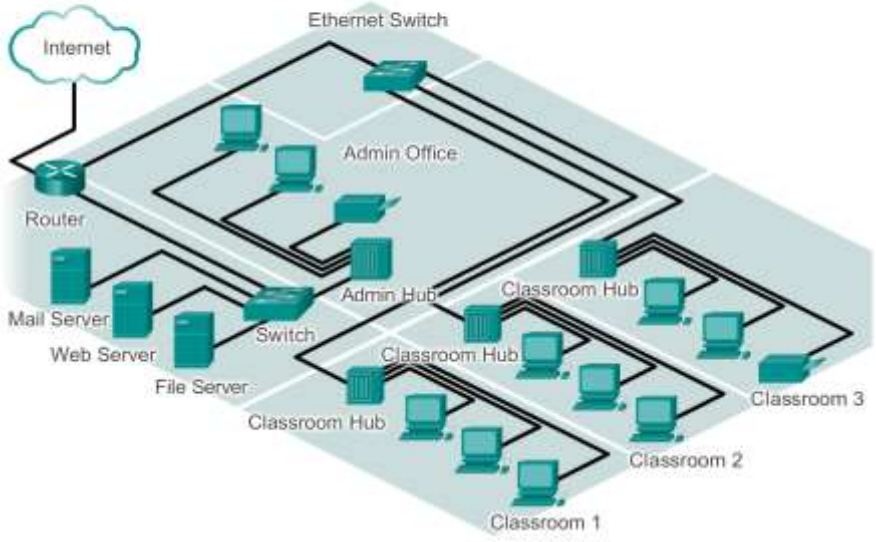
Network Representations



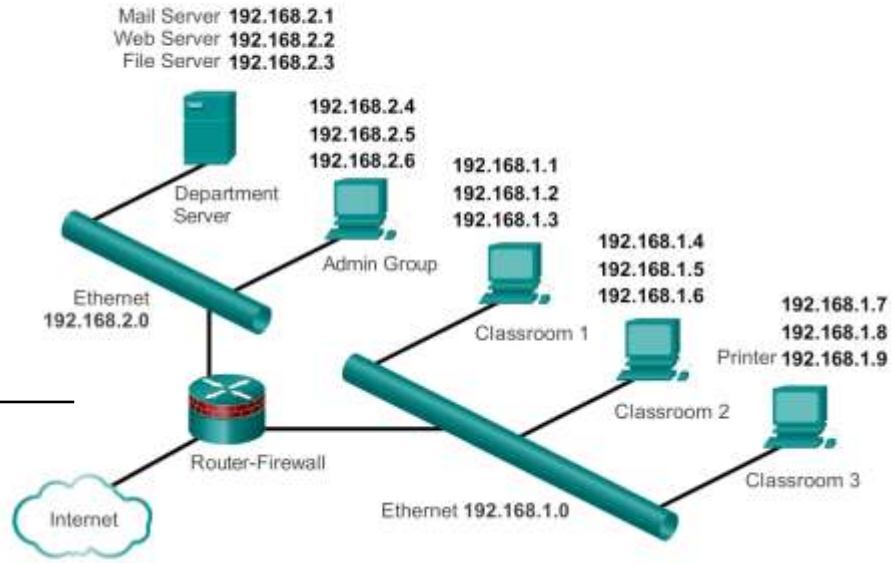
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Topology Diagrams

Physical Topology → physical location of intermediary devices, configured ports and cable installation



Logical Topology



identify devices, ports, IP addressing scheme ←

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Types of Networks

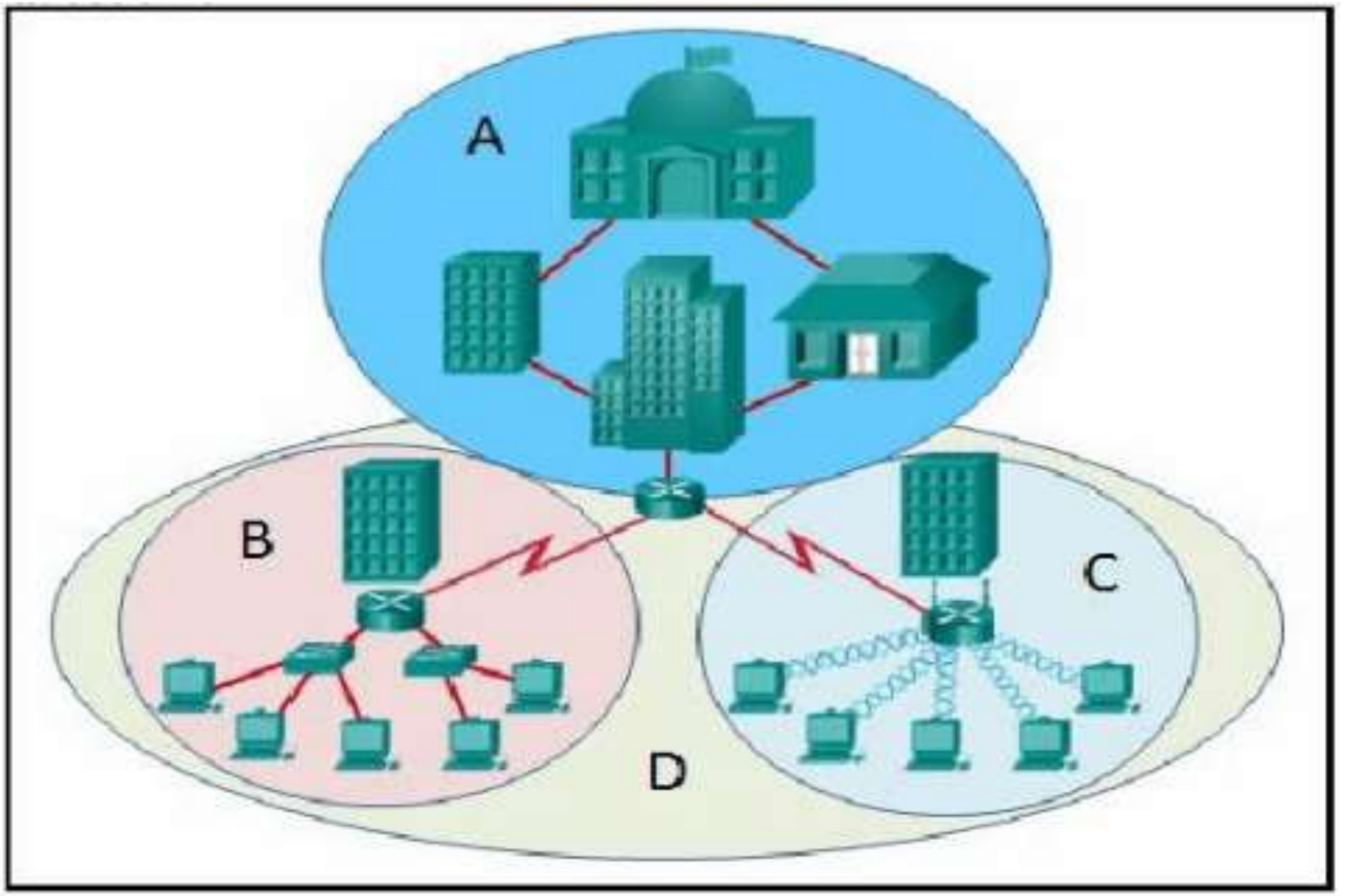
The two most common types of network infrastructures are:

- Local Area Network (LAN)
- Wide Area Network (WAN).

Other types of networks include:

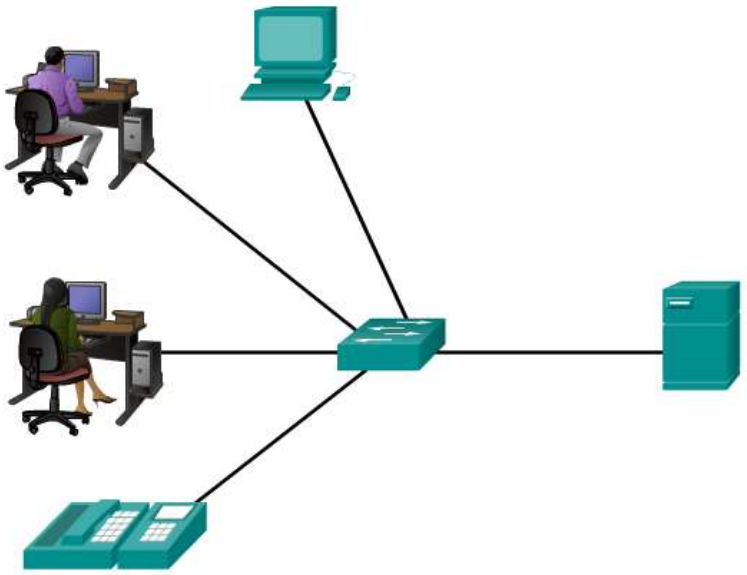
- Metropolitan Area Network (MAN)
- Wireless LAN (WLAN)
- Storage Area Network (SAN)

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Local Area Networks (LAN)



A network serving a home, building, or campus is considered a LAN.

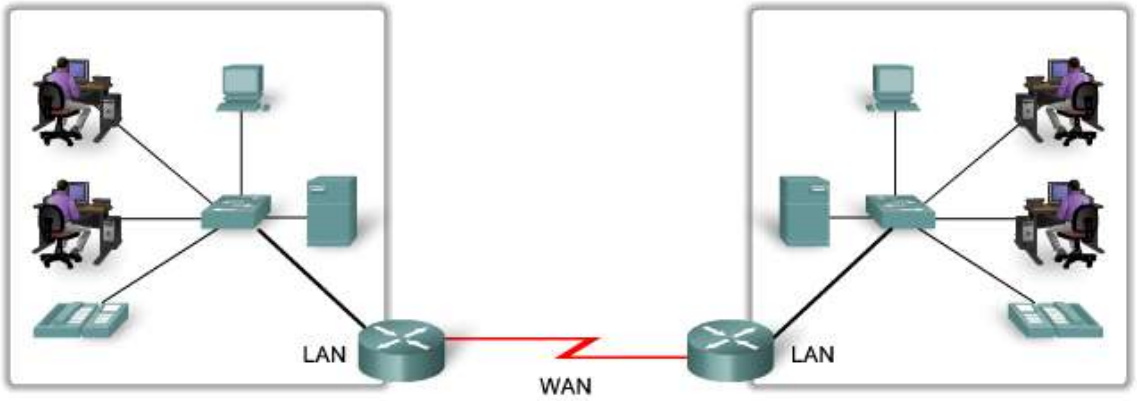
Specific features of LANs include:

- LANs interconnect end devices in a limited area such as a home, school, office building, or campus.
- A LAN is usually administered by a single organization or individual.
- LANs provide high speed bandwidth to internal end devices and intermediary devices.

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Wide Area Networks (WAN)

LANs separated by geographic distance are connected by a network known as a Wide Area Network (WAN).



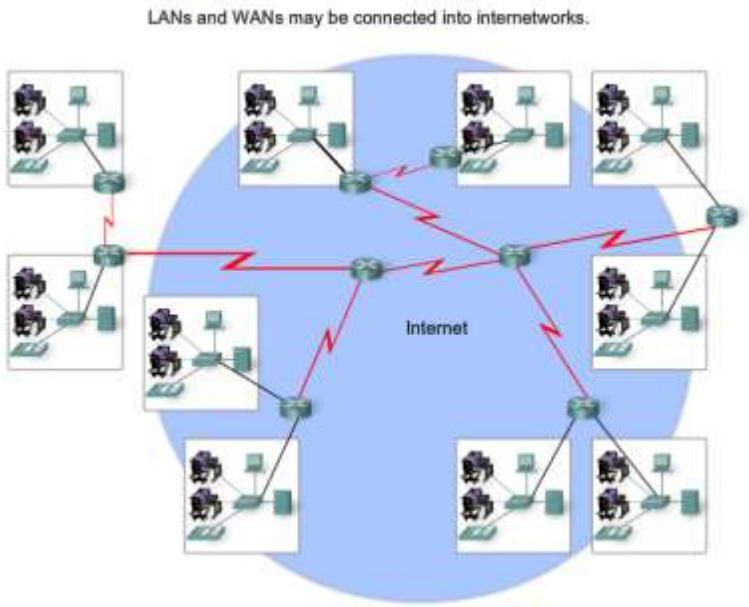
Specific features of WANs include:

- WANs interconnect LANs over wide geographical areas such as between cities, states, provinces, countries, or continents.
- WANs are usually administered by multiple service providers.
- WANs typically provide slower speed links between LANs.

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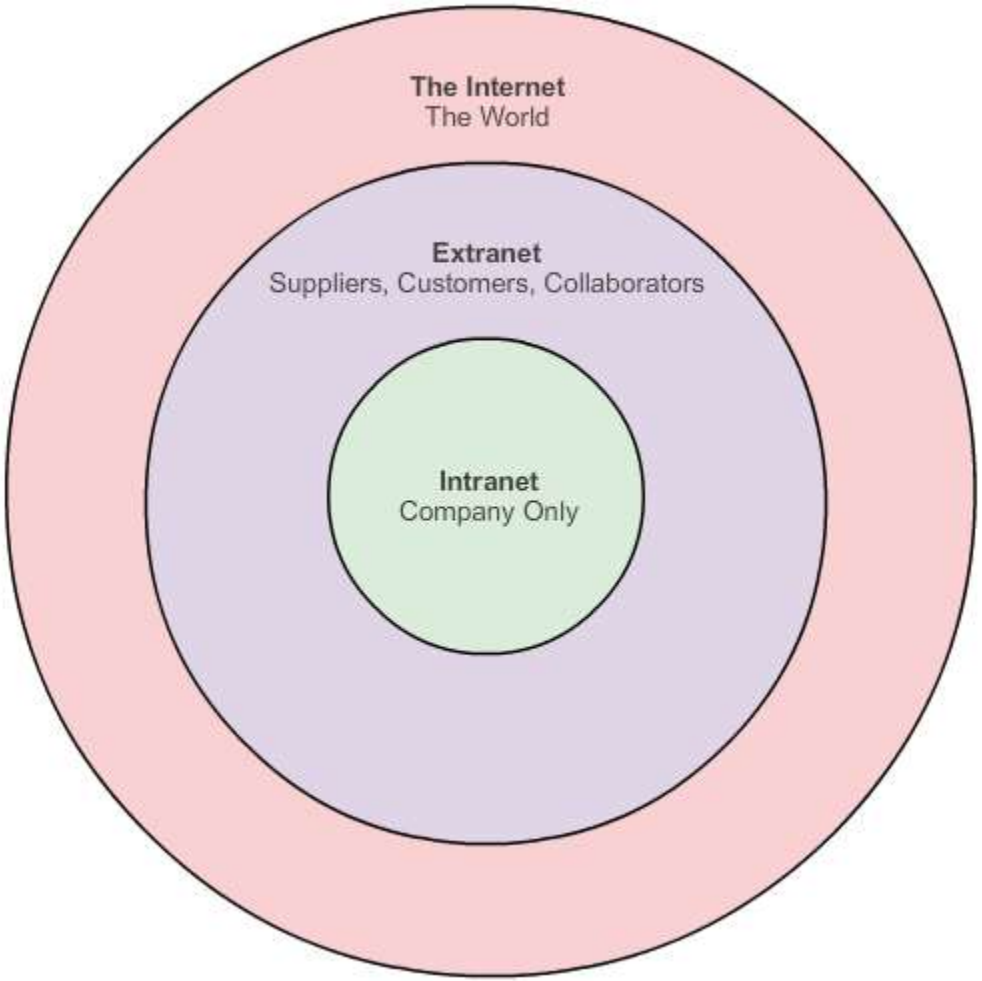
The Internet

- The Internet is a conglomerate of networks and is not owned by any individual or group.
- Ensuring effective communication across this diverse infrastructure requires the application of consistent and commonly recognized technologies and standards.
- These organizations include the Internet Engineering Task Force (IETF), Internet Corporation for Assigned Names and Numbers (ICANN), and the Internet Architecture Board (IAB), plus many others.



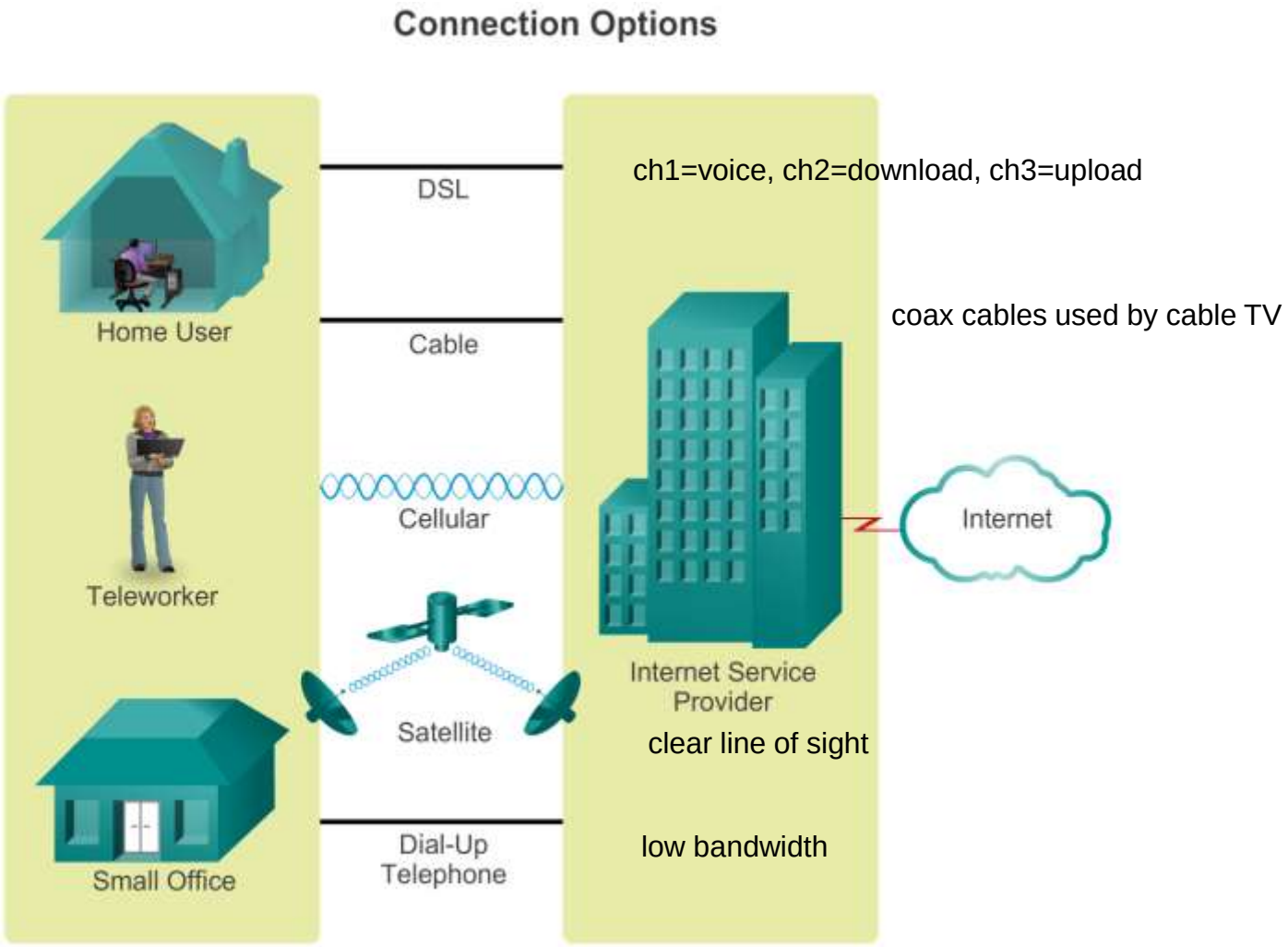
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Intranet and Extranet



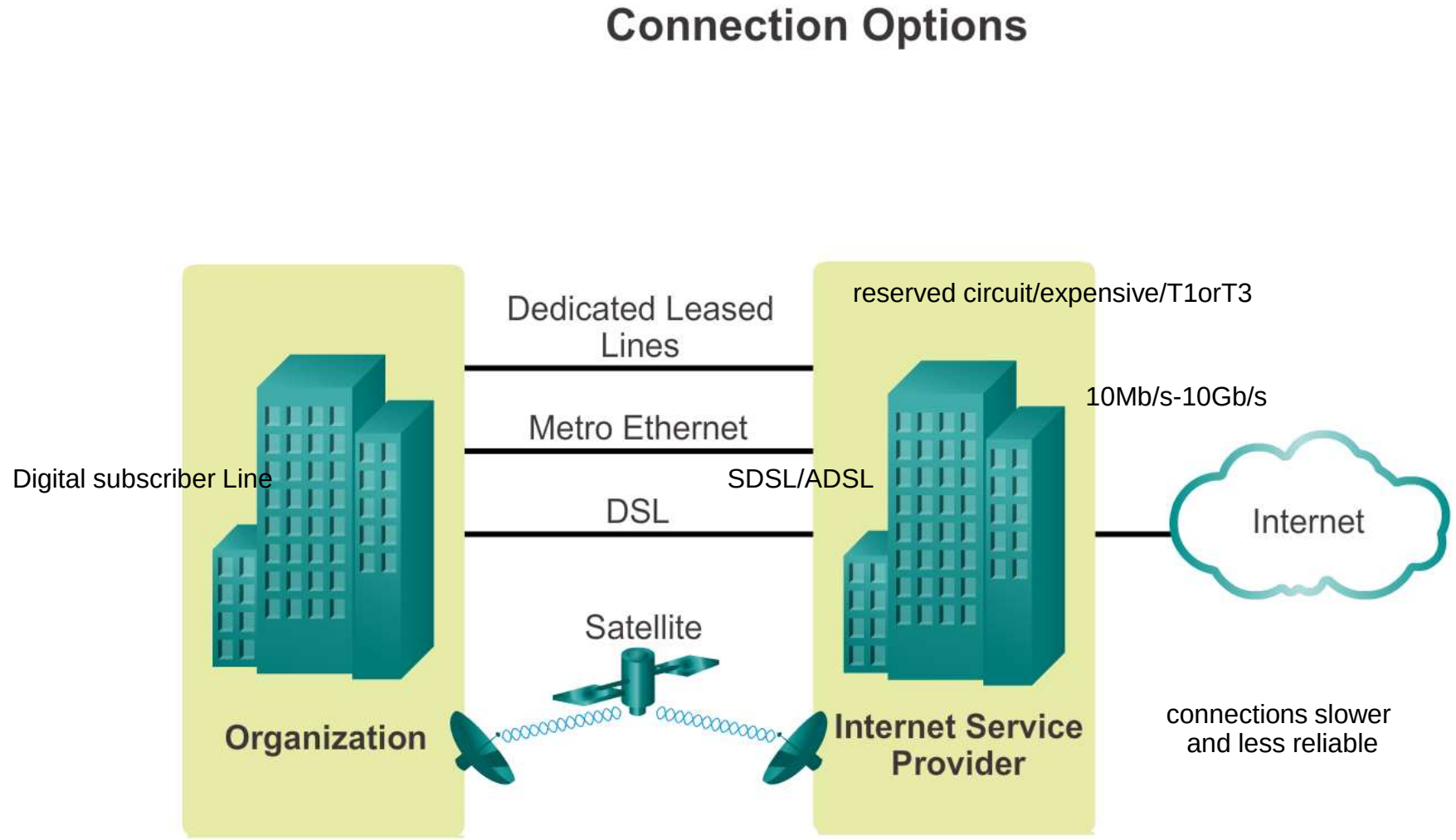
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Connecting Remote Users to the Internet



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Connecting Businesses to the Internet



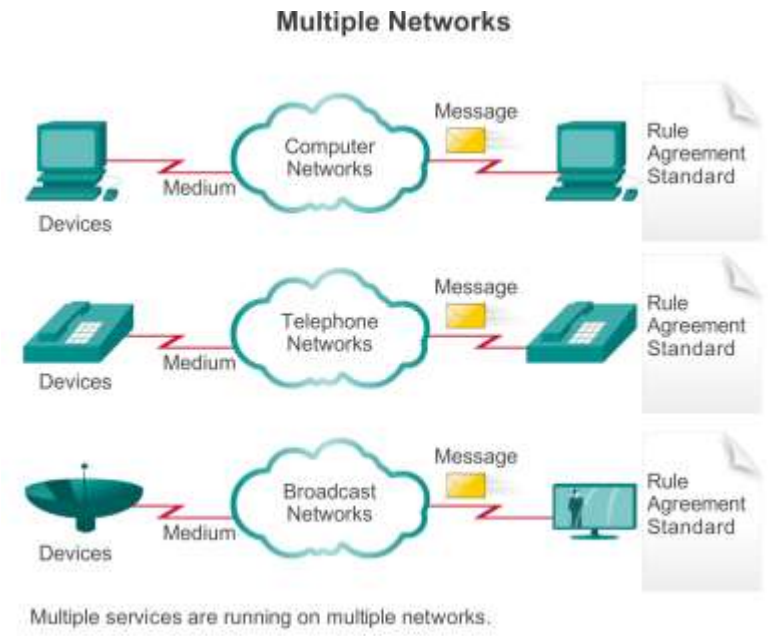
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1.3 The Network as a Platform

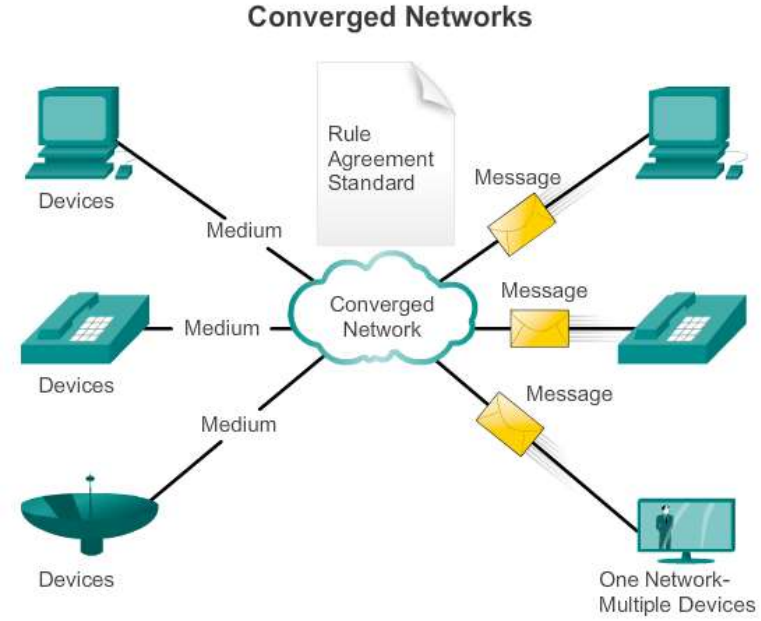


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The Converging Network



Multiple services are running on multiple networks.



Converged data networks carry multiple services on one network.

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Supporting Network Architecture

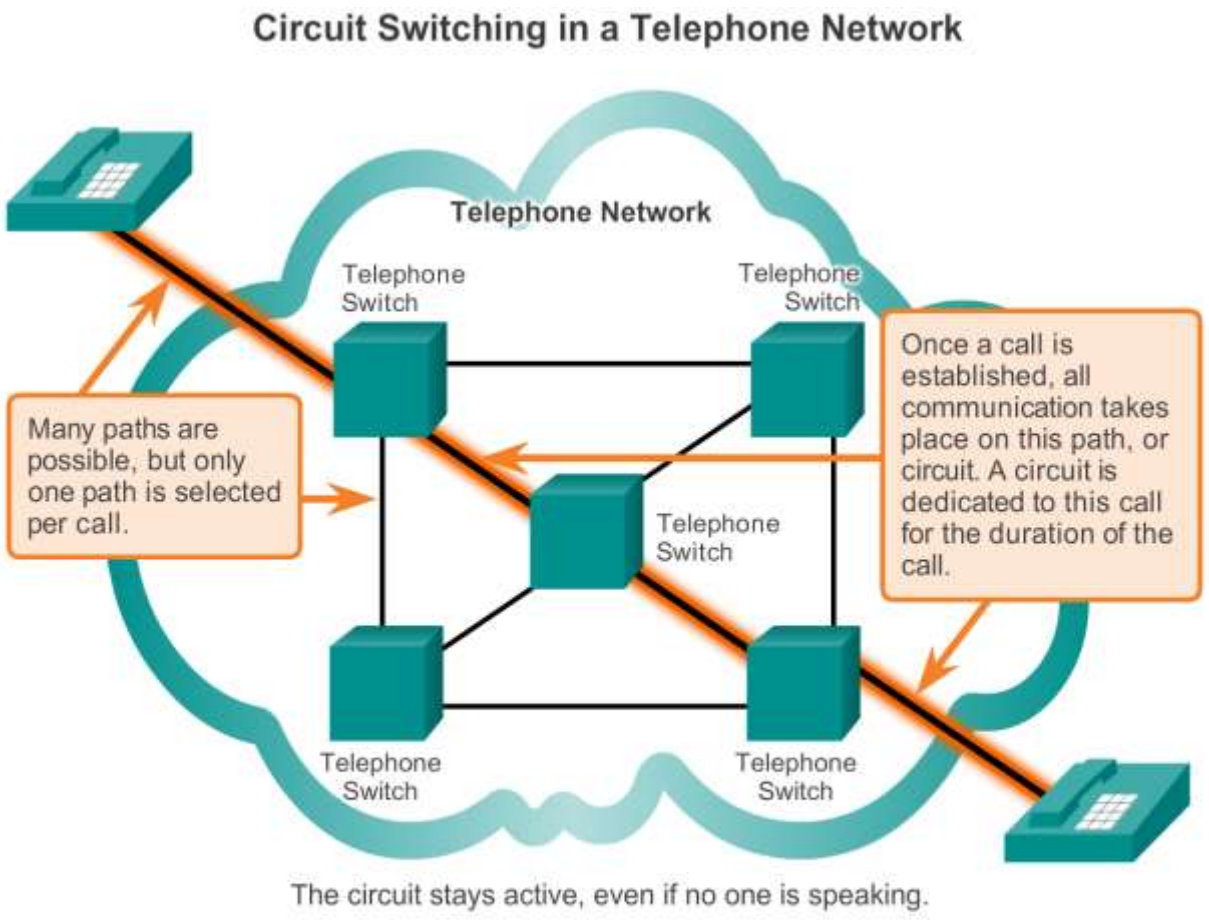
As networks evolve, we are discovering that there are four basic characteristics that the underlying architectures need to address in order to meet user expectations:

- Fault Tolerance (redundancy allows alternative routes)
- Scalability (add extra users without performance degrading)
- Quality of Service (QoS) (package priorities match type of communication)
- Security (software and hardware security)

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Reliable Network

Fault Tolerance in Circuit Switched Network

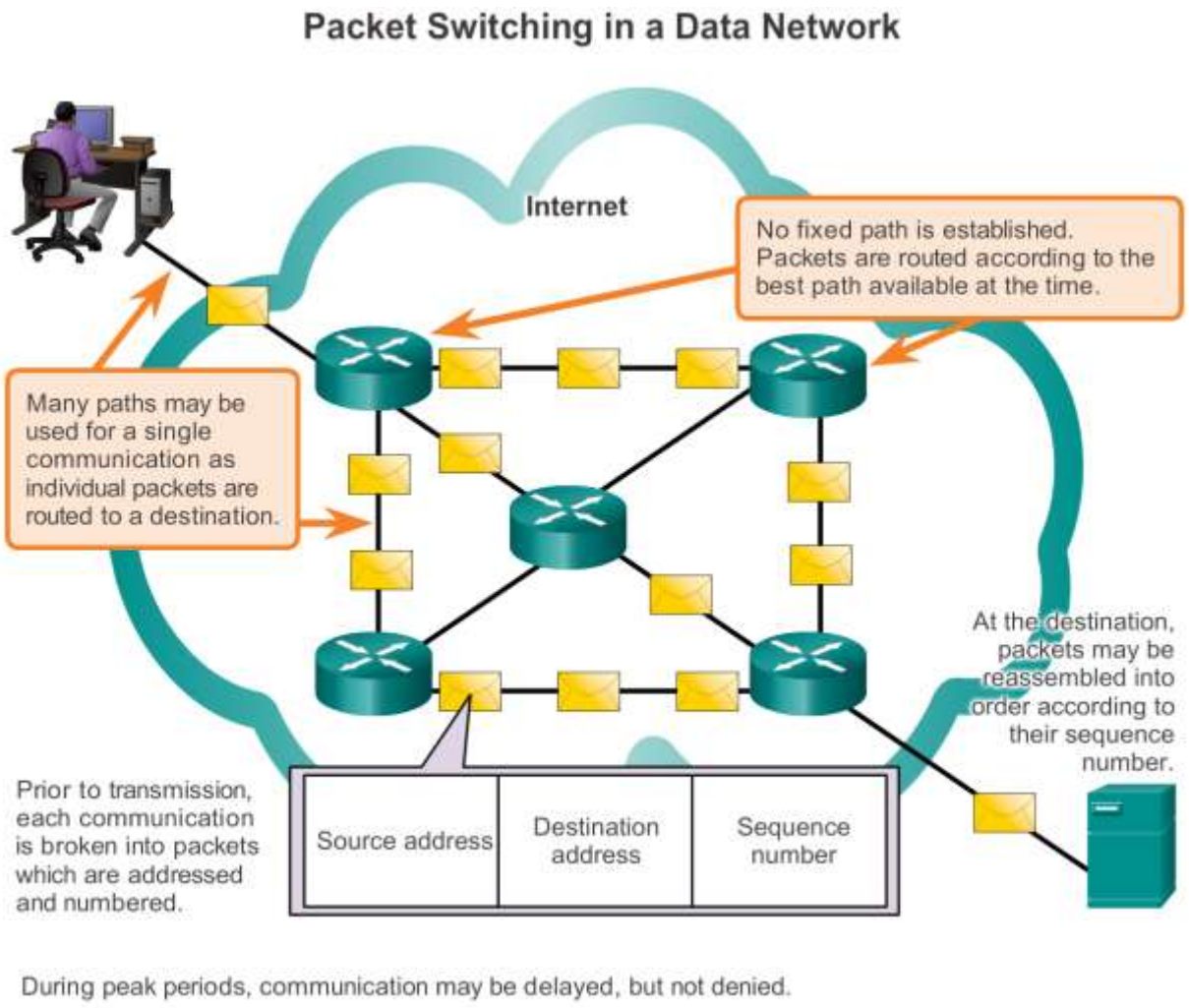


There are many, many circuits, but a finite number. During peak periods, some calls may be denied.

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Reliable Network

Packet-Switched Networks



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Reliable Network

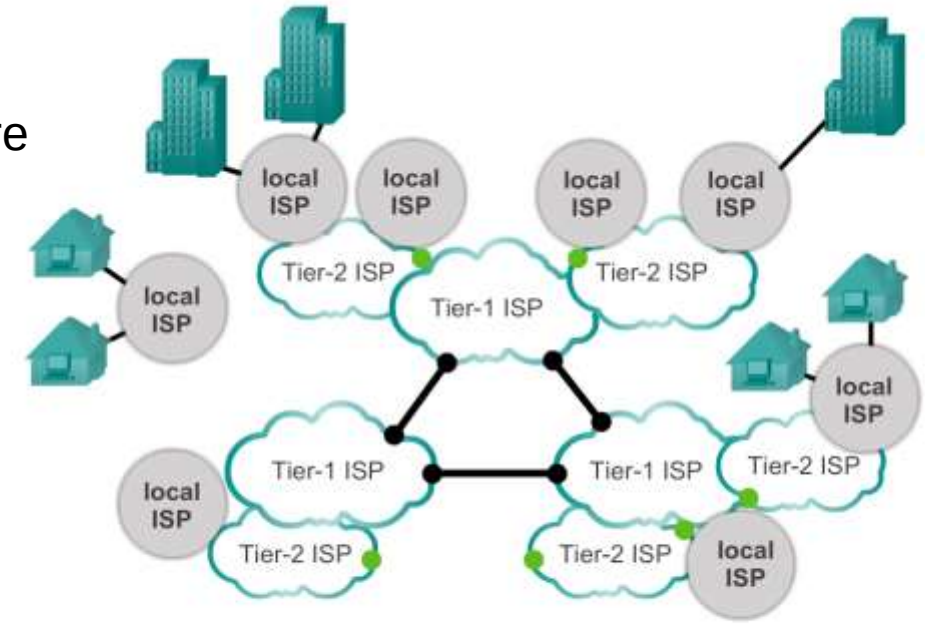
Scalable Networks

At the center of the Internet, Tier-1 ISPs provide national and international connections. These ISPs treat each other as equals.

Tier-2 ISPs are smaller and often provide regional service. Tier-2 ISPs usually pay Tier-1 ISPs for connectivity to rest of the Internet.

Peer connections between networks at the same level provide direct connections, bypassing longer routes and preventing congestion on the backbone.

layered structure



Tier-3 ISPs are the local providers of service directly to end users. Tier-3 ISPs are usually connected to Tier 2 ISPs and pay Tier 2 providers for Internet access.

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Providing QoS

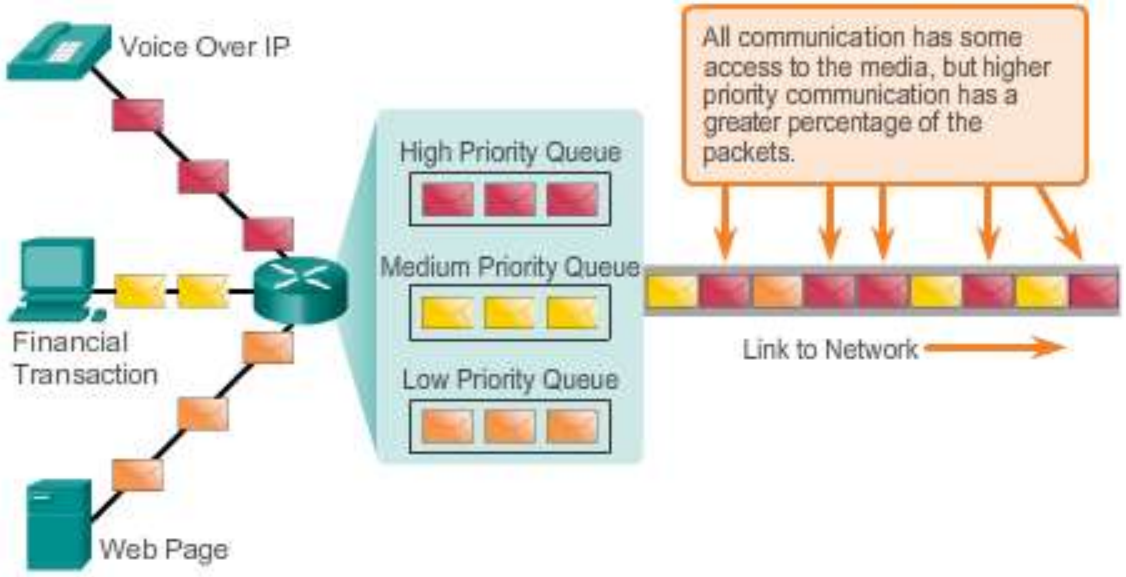
Examples of priority decisions for an organization might include:

- Time-sensitive communication - increase priority for services like telephony or video distribution.
- Non time-sensitive communication - decrease priority for web page retrieval or email.
- High importance to organization - increase priority for production control or business transaction data.
- Undesirable communication - decrease priority or block unwanted activity, like peer-to-peer file sharing or live entertainment.

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Providing QoS

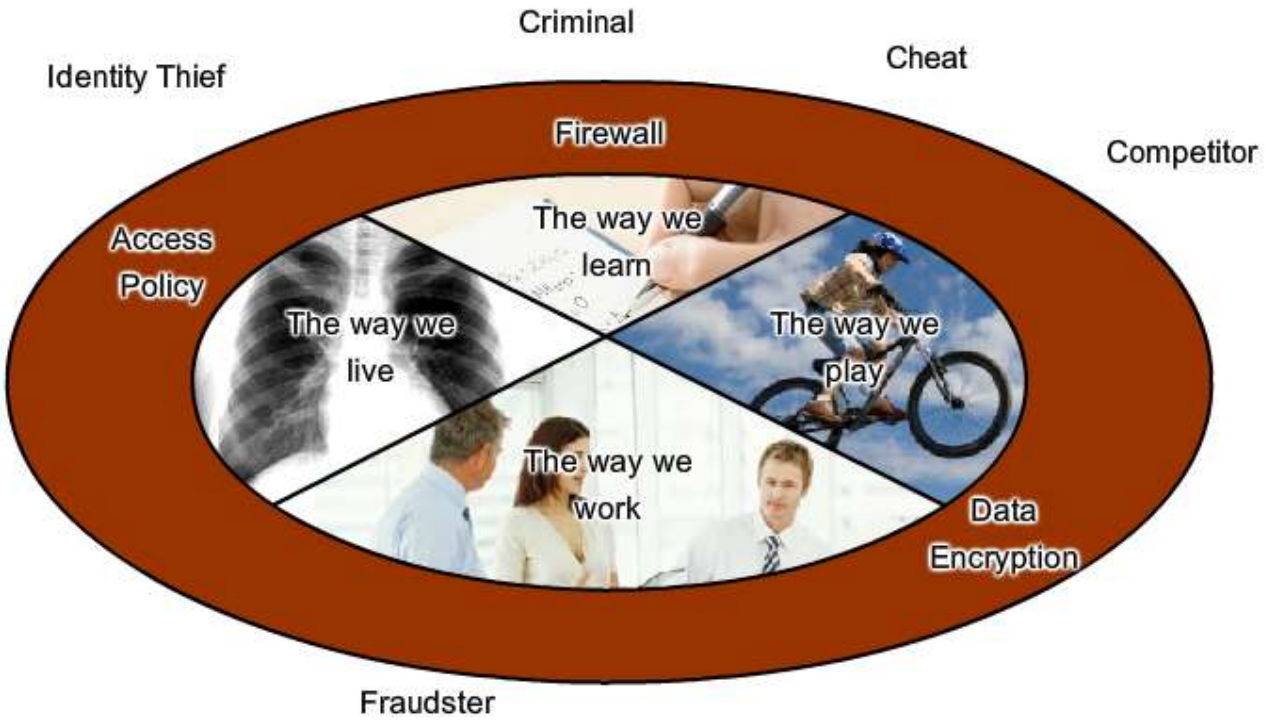
Using Queues to Prioritize Communication



Queuing according to data type enables voice data to have priority over transaction data, which has priority over web data.

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Providing Network Security



The communication and information that we would like to be private is protected from those who would make unauthorized use of it.

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Providing Network Security

There are two types of network security concerns that must be addressed:

- network infrastructure security - physical securing of devices and preventing unauthorized access to the management software that resides on them.
- information security - protecting the information contained within the packets being transmitted over the network and the information stored on network attached devices

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Providing Network Security

In order to achieve network security, there are three primary requirements:

- **Ensuring confidentiality** - only the intended and authorized recipients - individuals, processes, or devices can access and read data. This is accomplished by having passwords and encrypting data..
- **Maintaining communication integrity** - having the assurance that the information has not been altered in transmission, from origin to destination.
- **Ensuring availability** - having the assurance of timely and reliable access to data services for authorized users. Network firewall devices, along with desktop and server antivirus software can ensure system reliability.

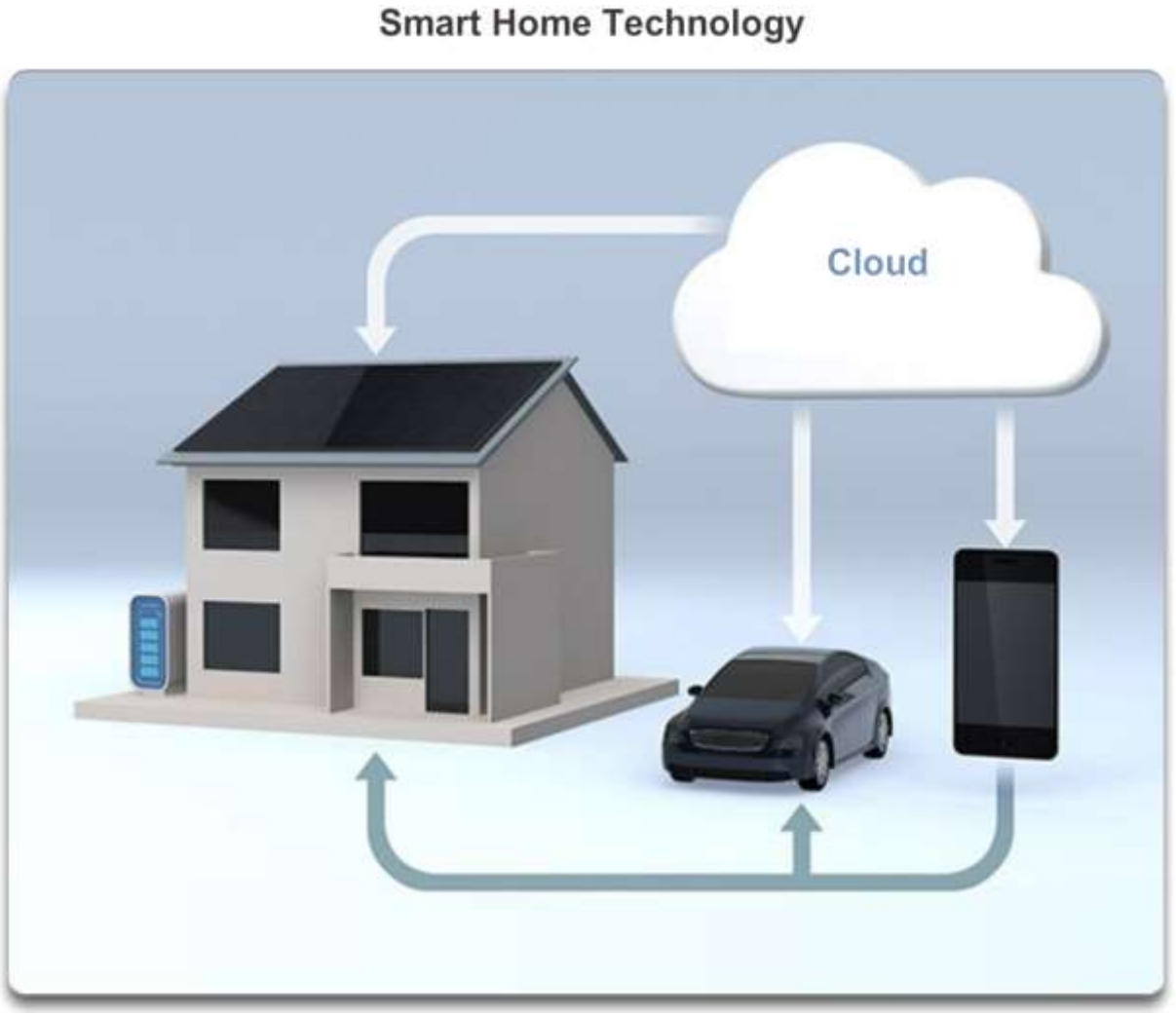
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1.4 The Changing Network Environment



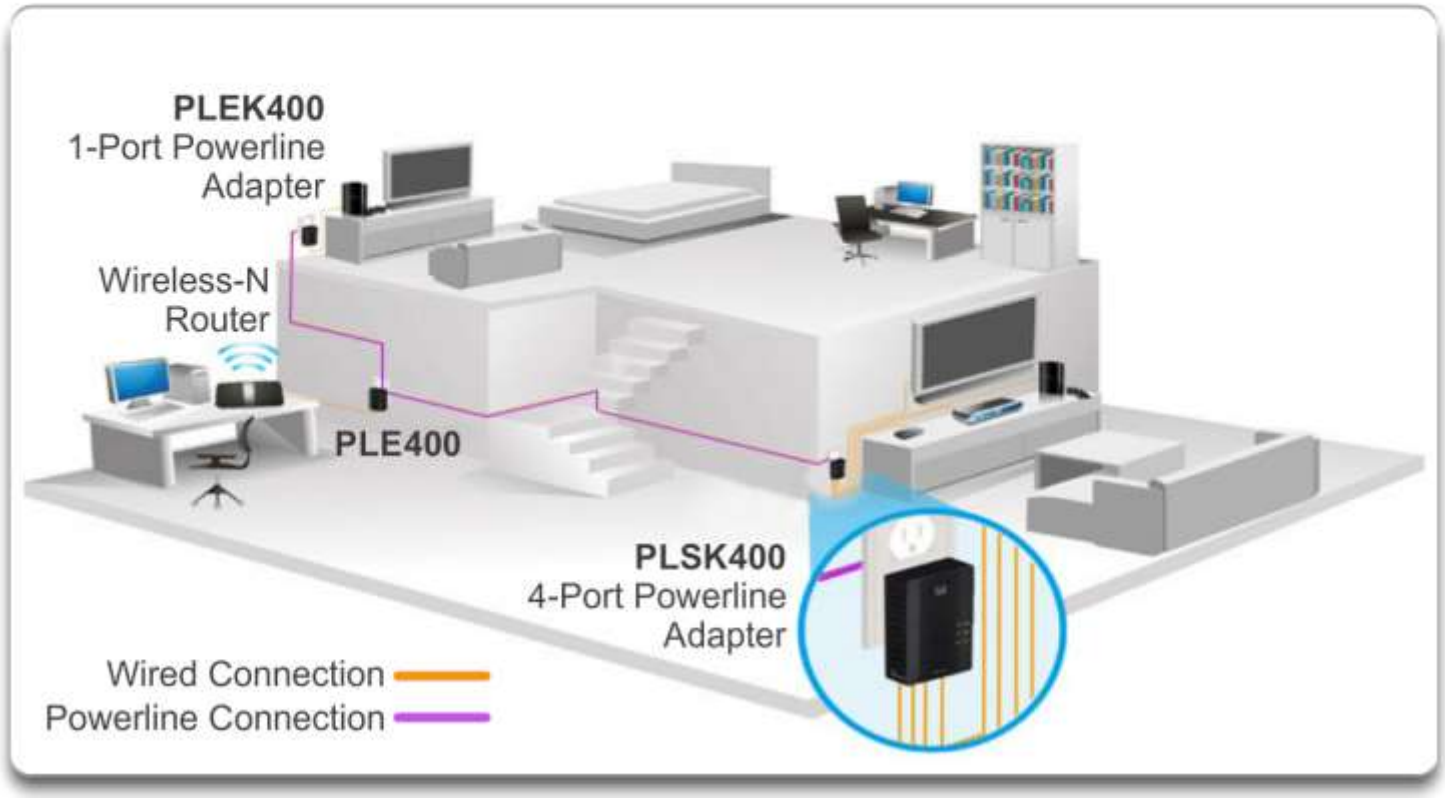
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Technology Trends in the Home



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Powerline Networking



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Wireless Broadband

Wireless Internet Service Provider (WISP)

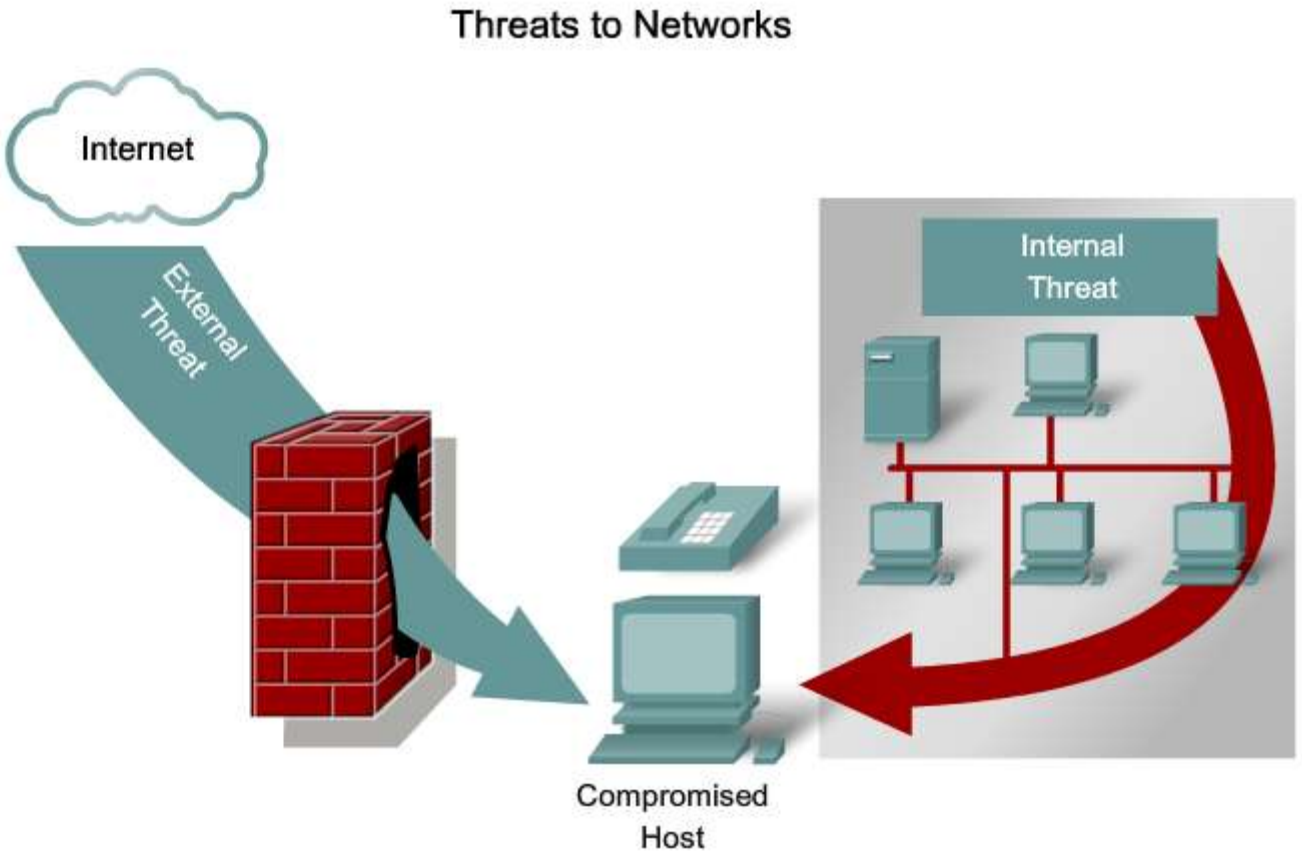
- Commonly found in rural environments where DSL or cable services are not available.
- A small dish or antenna is installed on the subscriber's roof in range of the WISP transmitter. The connection between the antenna and the WISP is wireless.

Wireless Broadband Service

- It uses the same technology that cellular technologies like smart phone and tablets are using. An antenna is installed outside the house providing either wireless or wired connectivity for devices in the home. In many areas, home wireless broadband is competing directly with DSL and cable services.

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Network Security



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Security Threats

The most common external threats to networks include:

- Viruses, worms, and Trojan horses
- Spyware and adware
- Zero-day attacks, also called zero-hour attacks
- Hacker attacks
- Denial of service (DoS) attacks
- Data interception and theft
- Identity theft

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Security Solutions

Network security components often include:

- Antivirus and antispyware
- Firewall filtering
- Dedicated firewall systems (more advanced system)
- Access control lists (ACL)
- Intrusion Prevention Systems (IPS)
- Virtual Private Networks (VPNs) (secure access to remote users)

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Summary

In this chapter, you learned:

- Networks and the Internet have changed the way we communicate, learn, work, and even play.
- Networks come in all sizes. They can range from simple networks consisting of two computers, to networks connecting millions of devices.
- The Internet is the largest network in existence. In fact, the term Internet means a ‘network of networks. The Internet provides the services that enable us to connect and communicate with our families, friends, work, and interests.
- The network infrastructure is the platform that supports the network. It provides the stable and reliable channel over which communication can occur. It is made up of network components including end devices, intermediate devices, and network media.

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Summary (cont.)

In this chapter, you learned:

- Networks must be reliable.
- Network security is an integral part of computer networking, regardless of whether the network is limited to a home environment with a single connection to the Internet, or as large as a corporation with thousands of users.
- The network infrastructure can vary greatly in terms of size, number of users, and number and types of services that are supported on it. The network infrastructure must grow and adjust to support the way the network is used. The routing and switching platform is the foundation of any network infrastructure.

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